Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Arctic Production Center
ARCTIC_ANALYSISFORECAST_BGC_002_004

Issue: 3.4

Contributors: Arne Melsom, Veli Çağlar Yumruktepe, Annette Samuelsen, Shuang Gao

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CHANGE RECORD

When the quality of the products changes, the QuID is updated, and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0		all	First version of document	Arne Melsom, Annette Samuelsen, Magne Simonsen	Laurent Bertino
1.1	May 2015	all	Change format to fit CMEMS graphical rules		L. Crosnier
1.2	January 2016		Update to reflect changes in observational product and temporal extension of validation results	Arne Melsom	
1.3	April 2016		Standard names of metrics	Laurent Bertino	
2.0	January 2017		Description of ECOSMO	Annette Samuelsen	Laurent Bertino
3.0	January 2021	All	New template, new system	Arne Melsom, Çağlar Yumruktepe	Laurent Bertino, ARC MFC leader
3.1	August 2022	All	Updated chlorophyll assimilation routine	Çağlar Yumruktepe, Arne Melsom	Laurent. Bertino, ARC MFC leader
3.2	June 2023	All	Added coccolithophores, CaCO3, variable detritus sinking speed as new model state variables and applied parameter tuning	Veli Çağlar Yumruktepe, Arne Melsom	Laurent Bertino, ARC MFC leader
3.3	May 2024	V, VI	Added a module for algae growth at the base of the sea ice	Annette Samuelsen, Shuang Gao	Laurent Bertino, ARC MFC leader
3.4	November 2025	V, VI	Advection of nutrients and algae in sea ice.	Shuang Gao, Veli Çağlar Yumruktepe, Arne Melsom	Laurent Bertino, ARC MFC leader

ACRONYM TABLE

Acronym	Description
CO_2	Carbon dioxide
L4	Level 4 processing: Gap-filling
RMSD	Root mean square difference
N/L, P/L, Si/L	Nitrate, phosphate, silicate per litre
Barents (Bar.)	Barents Sea. The region definition is displayed in Figure 1b.
NorwegianN (Nor. N.)	Norwegian Sea – north. The region definition is displayed in Figure 1b.
NorwegianS (Nor. S.)	Norwegian Sea – south. The region definition is displayed in Figure 1b.
IMR	Institute of Marine Research, Norway
WOA	World Ocean Atlas

KEY DEFINITIONS

Nudging	Nudging refers to an incremental adjustment for one product towards another product. In Section IV.1 satellite chlorophyll-a estimations were projected downward in the water column, by extrapolating surface concentrations using an empirical relation (Ardyna et al., 2013). The model chlorophyll-a profiles are then modified to achieve a closer value (30% of the difference) to the satellite estimated chlorophyll-a profiles.

RELEVANT DOCUMENT LIST

CMEMS ARC MFC PUM	Product User Manual for Arctic Ocean Physical and BGC Analysis and Forecasting Products[1]

[1] https://documentation.marine.copernicus.eu/PUM/CMEMS-ARC-PUM-002-ALL.pdf

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

The purpose of this document is to describe the quality of the Arctic biogeochemical near real-time and forecast product ARCTIC_ANALYSISFORECAST_BGC_002_004, run daily with weekly assimilation. The production system is described in Section II, and the validation framework is outlined in Section III. Validation results are detailed in six sub-sections of Section IV. Information about the history of the product is given in Sections V and VI.

The product results are disseminated on a regular grid with a spatial resolution of 6.25 km and 40 vertical levels, covering the Arctic Ocean and adjacent ocean regions. They include daily and monthly mean fields for the following 13 biogeochemical variables: net primary production, chlorophyll concentration, attenuation coefficient of light in sea water, nitrate concentration, phosphate concentration, phytoplankton biomass, zoo plankton biomass, dissolved oxygen concentration, silicate concentration, flux of sinking particulate organic matter, sea water pH, dissolved inorganic carbon concentration, and surface partial pressure of CO_2 in sea water.

I.2 Summary of the results

The quality of chlorophyll-a, nitrate, silicate, phosphate, surface CO_2 flux, and surface dissolved inorganic carbon was evaluated by comparing model results with observational data. The simulation was initialized from an observation based climatological state on 1 January 2013. The initial two-year period (2013 – 2014) was designated as the model spin-up, while the model output for the years 2015 and 2016 was compared to the *in situ* inorganic nutrient data (nitrate, silicate, and phosphate - from the Institute of Marine Research: Næringssalt-, oksygen- og klorofyll- data i norske havområder fra 1980–2017; IMR, 2018). The output data for the years 2015 – 2019 were compared to monthly averaged reprocessed gap-filled full domain (L4) product data[2] for surface CO_2 flux and CO_2 dissolved inorganic carbon validation. These years were selected to ensure consistency with the evaluation periods used in previous versions of this document. A parallel evaluation for chlorophyll-a was conducted using remote sensing data. We include comparisons between the quality of the current model version and the previous version from 2022. In this context, the previous version is referred to as ARC BGC22, while the current version is referred to as ARC BGC23.

Chlorophyll-a: The quality of the model results for the concentration of chlorophyll-a was evaluated using $\log_{10}$-transformed values in mg/m^3. For evaluations based on remote sensing data, model chlorophyll-a was integrated over the Atlantic sector region (see Section III), which encompasses the most ice-free regions. This yields a time-averaged surface bias of approximately 0.1 $\log_{10}$ (mg/m^3), indicating a surplus of chlorophyll-a in the model results. The bias is larger in the Barents Sea, reaching an average of 0.3 $\log_{10}$ (mg/m^3). The root mean square differences (RMSD) between observations and model results for the Atlantic sector are approximately 0.7 $\log_{10}$(mg/m^3). The discrepancies are largest during April, with biases in the 0.5-1 $\log_{10}$ (mg/m^3) range, and RMSD values of approximately 1 $\log_{10}$ (mg/m^3). These maxima coincide with the spring bloom.

In parallel, the quality of the model results for chlorophyll-a concentrations is evaluated by co-locating the model and *in situ* sampling points and interpolating the co-located model data to the corresponding

[2] https://doi.org/10.48670/moi-00047

sampling depths. The resulting biases range from 0.29 – 0.42 $\log_{10}$(mg/m^3), while the corresponding RMSD between observations and model results are between 0.65 – 0.79 $\log_{10}$ (mg/m^3).

Nitrate: The quality of model results for nitrate concentration is assessed after co-locating the model and *in situ* sampling points (IMR2018) and interpolating the co-located model data to the sampling depths. Model values are converted from µg C/L to µmol N/L. Biases range from -1.19 to -1.71 µmol N/L. Corresponding RMSDs between observations and model results range from 2.47 – 2.83 µmol N/L.

Phosphate: A validation approach similar to that used for nitrate was used for phosphate. Model values are converted from µg C/L to µmol P/L. Biases range from -0.1 to -0.13 µmol P/L. The corresponding RMSD between observations and model results are between 0.21 and 0.36 µmol P/L.

Silicate: After applying the same co-location and interpolation as for nitrate, model silicate values are converted from µg C/L to µmol Si/L. The resulting biases range from 0.28 to 0.63 µmol Si/L. The corresponding RMSDs between observations and model range from 1.39 to 1.66 µmol Si/L.

Surface dissolved inorganic carbon: The quality of the model results for the concentration of surface dissolved inorganic carbon is assessed by taking the model monthly and regional means (Figure 1b) and comparing them to the reprocessed L4 monthly and regional means[3] for the years between 2015 and 2019. The bias ranged from 13.15 to 57.02 mmolC m^{-3}, while the RMSD ranged from 17.23 to 61.72 mmolC m^{-3}. It should be noted that these statistics were calculated from 60 data points.

Surface-downward CO2 flux: A similar approach to surface dissolved inorganic carbon (using reprocessed L4 means) was employed for the statistical analysis of the surface downward flux of CO_2. The bias range was between 3.75 and 6.49 molC m^{-2} y^{-1}, while the RMSD range was from 4.6 to 7.3 molC m^{-2} y^{-1}. It should be noted that these statistics were calculated using 60 data points.

Primary production: Insufficient data (< 50 data points).

Phytoplankton: Insufficient data.

Zooplankton: Insufficient data.

Oxygen: Insufficient data.

Radiative attenuation: Insufficient data.

pH: Insufficient data.

Important notice: The observational coverage is too sparse to allow for a comprehensive assessment of the quality of model forecasts. Even chlorophyll-a, the most extensively observed biogeochemical variable, is subject to significant uncertainties due to limitations in coverage caused by persistent and extended cloud cover at high latitudes, and the absence of data during winter, when the sun does not rise sufficiently high above the horizon. In situ observations for nitrate, silicate, and phosphate are not available for recent years. Therefore, the years 2015 and 2016 were selected for nutrient validations, as they have a relatively high number of available sample points.

[3] https://doi.org/10.48670/moi-00047

I.3 Estimated Accuracy Numbers

This section provides estimated accuracy numbers (*Table 1*) based on the comparison between model results for the following variables: chlorophyll-a (year 2019), nitrate, silicate, phosphate (years 2015 and 201), and CO_2 (period 2015 – 2019) compared to observations from the sources that are listed in *Table 3* in Section III.

Table 1: Estimated accuracy numbers based on model results for 2019 for chlorophyll-a and 2015 and 2016 for nitrate, silicate, phosphate, and 2015 – 2019 for CO_2 variables compared with observations from the sources that are listed in Table 3 in Section III. Days refer to forecast days. See Figure 1 for the location of regions.

Variable	Region	Units	Mean bias		RMSD	
			Day 1	Day 6	Day 1	Day 6
chlorophyll-a against satellite data	Atlantic sector	$log_{10}(mg/m^3)$	0.10	0.09	0.70	0.71
	Nordic Seas		0.32	0.32	0.61	0.62
	Barents Sea		0.09	0.10	0.60	0.61
chlorophyll-a against *in situ* bottle data (see Table 4 for details)	NorwegianN	$log_{10}(mg/m^3)$	0.32		0.72	
	NorwegianS		0.42		0.79	
	Barents		0.29		0.65	
Nitrate (see Table 5 for a detailed breakdown of these values)	NorwegianN	µmol N/L	-1.71		2.83	
	NorwegianS		-1.19		2.47	
	Barents		-1.25		2.47	
Phosphate (see Table 6 for detailed breakdown of these values)	NorwegianN	µmol P/L	-0.13		0.36	
	NorwegianS		-0.10		0.36	
	Barents		-0.12		0.21	
Silicate (see Table 7 for detailed breakdown of these values)	NorwegianN	µmol Si/L	0.28		1.39	
	NorwegianS		0.63		1.59	
	Barents		0.34		1.66	
Surface dissolved inorganic carbon	NorwegianN	$mmolC/m^3$	13.2		17.2	
	NorwegianS		28.8		30.6	
	Barents		57.0		61.7	
Surface downwards CO_2 flux	NorwegianN	$molC/m^2/y$	5.94		6.41	
	NorwegianS		6.49		7.03	
	Barents		3.75		4.60	

II PRODUCTION SYSTEM DESCRIPTION

Production centre´s name: Norwegian Meteorological Institute

Production system name: Arctic Ocean biogeochemistry (BGC) analysis and forecast system (Copernicus Marine Environment Monitoring Service (CMEMS) name: ARCTIC_ANALYSISFORECAST_BGC_002_004)

Description:

The Arctic Ocean biogeochemistry analysis and forecast system comprises the ocean general circulation Hybrid Coordinate Ocean Model (HYCOM; Chassignet et al., 2009) version 2.98, coupled with the ECOSystem MOdel (ECOSMO II(CHL); Yumruktepe et al., 2022).

The model system is configured using a grid with a horizontal resolution of 6 km and 45 vertical layers using a hybrid z/isopycnal coordinate system. The model results are interpolated onto a 6.25 km stereographic grid and 40 z-levels, which are distributed as ARCTIC_ANALYSISFORECAST_BGC_002_004 product.

The model is driven by surface ocean-atmosphere fluxes from the European Centre for Medium-Range Weather Forecasts (ECMWF). The simulations are conducted on daily basis, and the results for the daily averages are made available to users. All regions north of 50°N are included, except the western Bering Sea and the northern Pacific Ocean.

The weekly production cycle is outlined in Table 2. It should be noted that for the Thursday bulletin, observations from the preceding Tuesday are assimilated. For this purpose, we use data from the CMEMS product OCEANCOLOUR_GLO_BGC_L3_NRT_009_101 (https://doi.org/10.48670/moi-00278). The assimilation of chlorophyll-a data is performed using a nudging approach, in which surface observations are projected downward in the water column using the Ardyna2013 algorithm, proposed by Ardyna et al. (2013). The Ardyna2013 parameters for the Norwegian and Barents Sea post-bloom period were modified to obtain a better profile fit with respect to in situ data obtained from IMR. Different versions of the depth projection of surface values were applied for pre-bloom, bloom, post-bloom, and winter periods.

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Table 2: The weekly production cycle. "Bulletin Day" is the day on which a forecast is produced while "Valid day" refers to days for which forecasts are available. The letter 'A' indicates simulations with analysed atmospheric forcing fields, 'A' (on Bulletin Day) refers to a combination of analysed atmospheric forcing (up to and including 12 UTC) and forecasts, 'F' corresponds to production using atmospheric forecasts, and 'P' refers to the assimilation of the biogeochemical state*

		Valid day															
		Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu
Bulletin Day	Mo	*A**	*F*	*F*	*F*	*F*	*F*	*F*	*F*	*F*	*F*						
	Tu		*A**	*F*	*F*	*F*	*F*	*F*	*F*	*F*	*F*	*F*					
	We			*A**	*F*	*F*	*F*	*F*	*F*	*F*	*F*	*F*	*F*				
	Th			*P*	*A**	*F*	*F*	*F*	*F*	*F*	*F*	*F*	*F*	*F*			
	Fr					*A**	*F*	*F*	*F*	*F*	*F*	*F*	*F*	*F*	*F*		
	Sa						*A**	*F*	*F*	*F*	*F*	*F*	*F*	*F*	*F*	*F*	
	Su							*A**	*F*	*F*	*F*	*F*	*F*	*F*	*F*	*F*	*F*

III VALIDATION FRAMEWORK

The validation of the ARCTIC-ANALYSISFORECAST-BGC_002_004 product was based on the assessment of the variables: chlorophyll-a, nitrate, silicate, and phosphate, as shown in Table 3. Other model variables (e.g., plankton and individual functional type biomasses) were not available to offer conclusive validation statistics as was the case for nutrients and chlorophyll-a.

The model results were compared with the relevant observations using the MERSEA Class 4 diagnostics (Crosnier and Le Provost, 2006). When a sufficient quantity of daily updated data is available, it is incorporated into the model/data comparison. However, such a requirement is rarely met for biogeochemical variables. To address this limitation, we carry out comparisons between model results and climatological fields derived from observations and *in situ* samples from previous years.

The results for the available quantities include the biases and RMSD between the model results and the observations. Biases are calculated such that positive values indicate that the model overestimates the observations.

The model results for chlorophyll-a presented here are calculated based on a reprocessed CMEMS product for the years 2015 and 2016. Only results for the surface chlorophyll concentrations are available. The regions for which results are presented are depicted in Figure 1: a) Regions used to validate the chlorophyll-A results. Blue: Nordic Seas; black: Barents Sea; red-blue-black composite: Atlantic sector. b) Regions used for nitrate, silicate, and phosphate validation. Regions shown are Barents (Bar.), NorwegianN (Nor. N.) and NorwegianS (Nor. S.). a.

Model validation statistics for nitrate, silicate, and phosphate presented were calculated with reference to *in situ* samples for 2015 (IMR2018, individual points) and World Ocean Atlas 2013 monthly climatology (Garcia et al., 2014). The *in situ* samples cover a range of depth points. The regions for which results are presented are shown in Figure 1b.

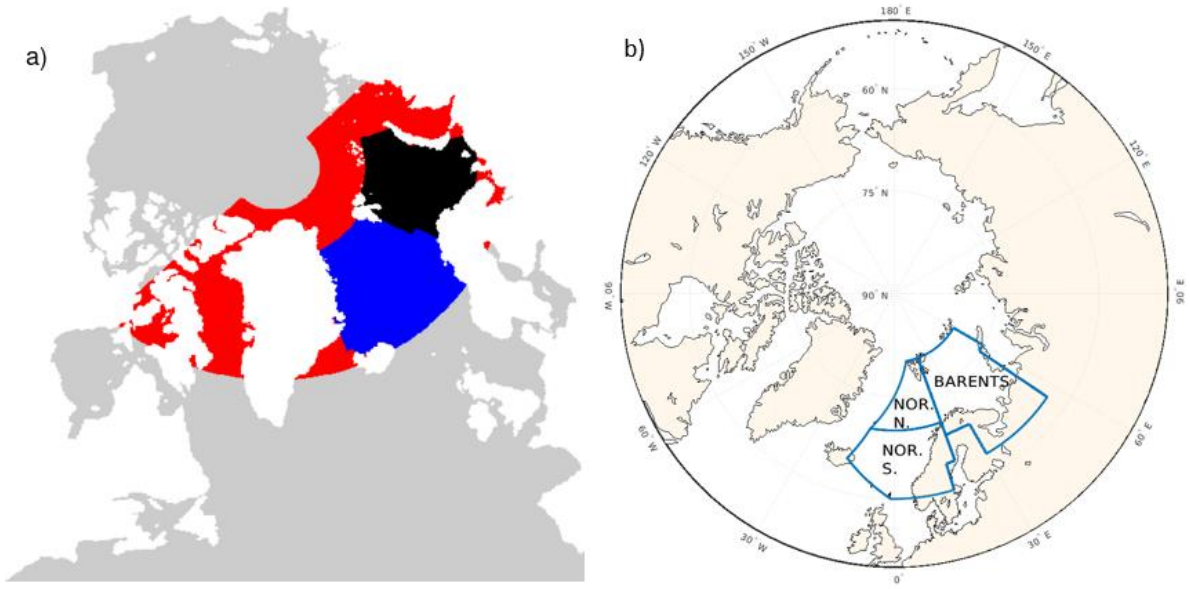

Figure 1: a) Regions used to validate the chlorophyll-A results. Blue: Nordic Seas; black: Barents Sea; red-blue-black composite: Atlantic sector. b) Regions used for nitrate, silicate, and phosphate validation. Regions shown are Barents (Bar.), NorwegianN (Nor. N.) and NorwegianS (Nor. S.).

Table 3: List of observational data sets, metrics, and diagnostics used in the validation of the ARCTIC-ANALYSISFORECAST-BGC_002_004 product.

Variable	Supporting Observations	Diagnostics
chlorophyll-a	- Data derived from satellite observations from 2015, 2016 in product OCEANCOLOUR_GLO_BGC_L3_MY_009_103 (multi sensors products based on SeaWIFS, MERIS, MODIS-A, MODIS-T, VIIRS-SNPP & JPPS1, OLCI-Sentinel-3A & Sentinel-3B) - In situ samples for 2015 and 2016 (IMR2018).	- Bias and RMSD for forecast ranges of 1 and 6 days, from mid-February to the end of October - Bias and RMSD when compared to in situ data; for the full year/ summer/ spring period
nitrate, silicate and phosphate	- In situ samples for 2015 and 2016 (IMR2018). - World Ocean Atlas 2013 monthly climatology.	- Bias, RMSD, full-year/ summer/ spring period
Surface CO_2 concentration and downward flux	- Monthly averaged L4 product MULTIOBS_GLO_BIO_CARBON_SURFACE_REP_015_008	- Bias, RMSD, full-year period

IV VALIDATION RESULTS

IV.1 Chlorophyll-a

The quality of the results for chlorophyll-a has been evaluated by comparison with the reprocessed daily data at a resolution of 4 km from the product OCEANCOLOUR_GLO_BGC_L3_MY_OBSERVATIONS_009_103. This observation product contains data from the MODIS, OLCI, and VIIRS instruments. Validation statistics (bias and RMSD) were calculated based on $\log_{10}$ transformed concentrations.

The observational product used for this the analysis of model results for 2015 and 2016 was assimilated by nudging. Furthermore, observations were also projected downward in the water column, by extrapolating surface concentrations using an empirical relation (Ardyna et al., 2013).

Day 1 forecast

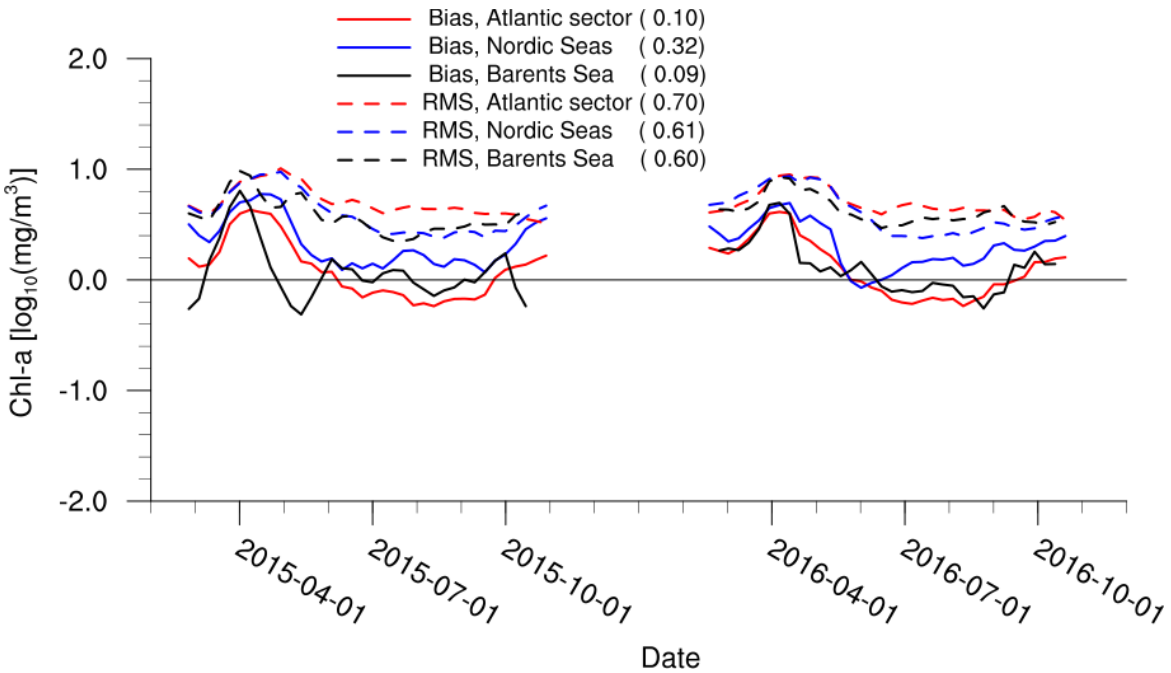

Figure 2: Time series of averaged chlorophyll-a bias and RMSD for the first forecast day in different regions (see Figure 1 for a detail on the regions). Numbers in brackets correspond to averages for the full-time series. Numbers along the vertical axis correspond to $\log_{10}$-transformed concentrations (in units of $\log_{10}(mg/m^3)$). The regions' colours match those in the map in Figure 1a (but here, red is for the combined red-blue-black regions in Figure 1).

The validation results for a 1-day forecast show that the model exhibits a positive overall bias for chlorophyll-a, mainly owing to the results in spring and autumn, but negative in summer (Figure 2). Overall, the bias is positive in all regions. RMSD is approximately 0.2 $\log_{10}(mg/m^3)$) units higher in spring but exhibits a generally low variability throughout the year.

Day 6 forecast

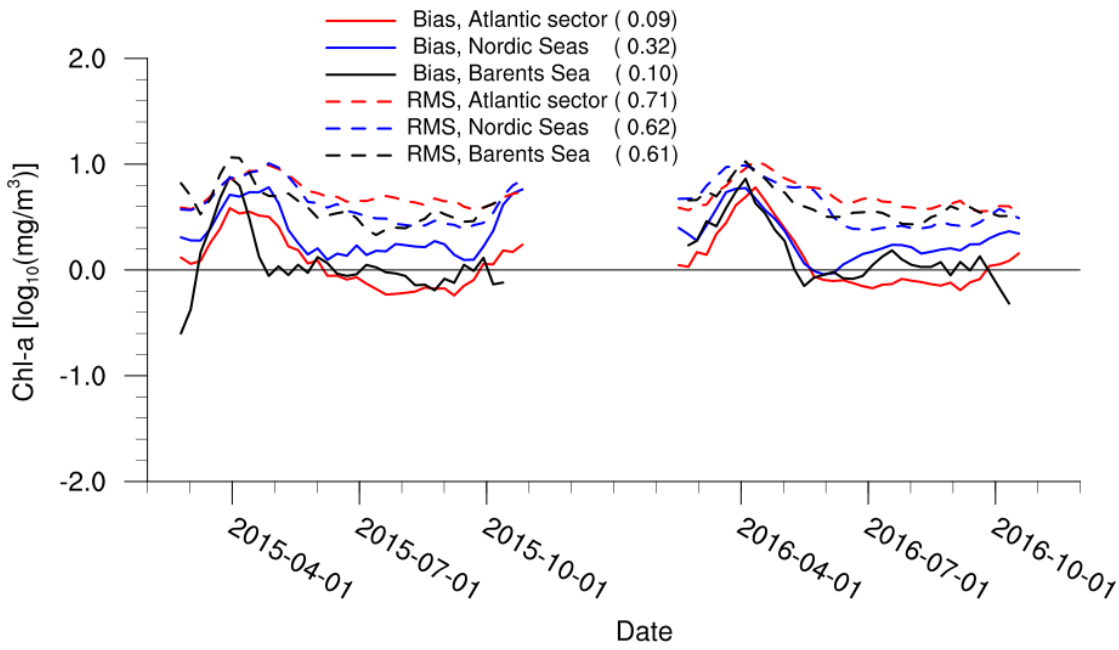

Figure 3: Time series of averaged chlorophyll-a bias and RMSD for the day 6 forecast in different regions (see Figure 1). Numbers in brackets correspond to averages for the full-time series. Numbers along the vertical axis correspond to log10-transformed concentrations (in units of log10(mg/m3)). The regions' colours match those in the map in Figure 1a (but here, red is for the combined red-blue-black regions in Figure 1).

The comparison with observational data for forecast day 1 (Figure 2) and forecast day 6 (

Figure 3) demonstrates that the quality of the data remains relatively consistent for all lead times.

Furthermore, model results are also compared to IMR2018 *in situ* profiles, after co-location of model and *in situ* sampling points and interpolation of model data at these co-located points onto the corresponding sampling depths. A breakdown of the statistics by season and depth range (Table 4) indicates no significant quality differences between ARC BGC22 (previous version of the product) and ARC BGC23 (current version). ARC BGC23 results in increased phytoplankton production during seasons with lower light availability and depth. This is reflected in the statistics where most of the improvements in ARC BGC23 are limited to these low-light conditions (see Section I.2 for the ARC BGC nomenclature). While the performance of the model chlorophyll-a improves with the use of surface chlorophyll-a assimilation, the degree of correction effectiveness is dependent on the cloud cover, which is high for the Arctic region (King et al., 2013).

Table 4: Estimated RMSD and bias for ARC BGC23 model predictions of chlorophyll-A log_{10}(mg/m3) for 2015 and 2016 compared to IMR2018 data. The RMSD for ARC BGC22 is shown in parentheses. The best value is highlighted in bold. In addition to the full-year metrics, model statistics are also provided for spring (April – June) and summer (July – September) periods and for different depth intervals with "0m – Bottom" covering the entire water column.

Depth (m)	Barents Full Year	Nor. N. Full Year	Nor. S. Full Year	Bar. Spring	Nor. N. Spring	Nor. S. Spring	Bar. Summer	Nor. N. Summer	Nor. S. Summer
RMSD, in log_{10}(mg/m^3)									
0 -10	0.402 **(0.34)**	0.683 **(0.565)**	0.764 **(0.543)**	0.552 **(0.302)**	0.716 **(0.676)**	0.7 **(0.566)**	0.386 **(0.352)**	0.41 **(0.347)**	**0.335** (0.377)
10 -20	0.414 **(0.35)**	0.607 **(0.537)**	0.705 **(0.471)**	0.672 **(0.38)**	**0.687** (0.736)	0.545 **(0.49)**	0.375 **(0.353)**	0.363 **(0.303)**	**0.447** (0.477)
20 -50	0.635 **(0.519)**	0.733 **(0.637)**	0.751 **(0.592)**	0.763 **(0.61)**	**0.784** (0.839)	0.65 **(0.616)**	0.623 **(0.514)**	0.614 **(0.418)**	0.883 **(0.692)**
50 -100	**0.906** (0.943)	0.807 **(0.685)**	0.811 **(0.663)**	0.716 **(0.632)**	**0.636** (0.739)	**0.596** (0.669)	**0.928** (0.982)	0.876 **(0.676)**	1.201 **(0.894)**
100 - 200	**0.915** (1.049)	0.693 **(0.461)**	0.852 **(0.696)**	0.663 **(0.588)**	**0.249** (0.456)	**0.81** (0.914)	**0.952** (1.066)	0.796 **(0.447)**	0.895 **(0.643)**
0 - Bottom	0.648 **(0.622)**	0.722 **(0.614)**	0.793 **(0.616)**	0.677 **(0.508)**	**0.715** (0.754)	0.656 **(0.617)**	0.649 **(0.64)**	0.621 **(0.467)**	0.857 **(0.693)**
BIAS, in log_{10}(mg/m^3)									
0 -10	**-0.063** (-0.119)	0.161 **(-0.006)**	0.282 **(-0.0)**	-0.294 **(-0.076)**	**-0.072** (0.411)	**-0.002** (0.376)	**-0.079** (-0.144)	**-0.077** (-0.197)	**-0.065** (-0.272)
10 -20	**0.002** (-0.095)	0.157 **(0.013)**	0.29 **(0.075)**	-0.273 **(-0.052)**	**0.001** (0.434)	**0.059** (0.319)	**-0.006** (-0.123)	**-0.044** (-0.138)	**-0.08** (-0.261)
20 - 50	0.332 **(0.201)**	0.368 **(0.215)**	0.329 **(0.062)**	**-0.03** (0.287)	**0.22** (0.554)	**0.022** (0.375)	0.345 **(0.185)**	0.351 **(0.168)**	0.471 **(0.253)**
50 -100	0.749 **(0.502)**	0.513 **(0.239)**	0.418 **(0.163)**	**-0.063** (0.252)	**0.099** (0.363)	**0.013** (0.403)	0.82 **(0.524)**	0.741 **(0.416)**	1.082 **(0.729)**
100 - 200	0.676 **(0.667)**	0.559 **(0.046)**	0.699 **(0.103)**	-0.613 **(-0.44)**	**-0.162** (0.44)	**0.519** (0.622)	**0.814** (0.88)	0.759 **(0.228)**	0.842 **(0.573)**
0 - Bottom	0.29 **(0.154)**	0.318 **(0.125)**	0.42 **(0.03)**	-0.163 **(0.107)**	**0.074** (0.448)	**0.054** (0.375)	0.311 **(0.147)**	0.278 **(0.08)**	0.529 **(0.18)**

IV.2 Nitrate

The quality assessment for nitrate is based on a comparison with available *in situ* data for the years 2015 and 2016 (Figure 4) due to the abundance of samples collected during this period and to facilitate intercomparisons with previous model quality assessments. Table 5 provides a detailed breakdown of the summary statistics presented in Table 1, organized by season and depth interval. The nitrate observations retrieved from IMR2018 were not incorporated into the model. The simulated years are part of a hindcast that ultimately merges with the operational run. Additionally, the model results for 2015 to 2019 were averaged on a regional basis (as shown in Figure 1b) and monthly for comparison with the monthly averages of the World Ocean Atlas 2013 (WOA2013) (Garcia et al., 2014) for the same region. The modelled nitrate concentrations are in good overall agreement with observations (Figure 5). Especially at the surface (5 m), the model can represent the seasonal decrease in nutrient concentrations due to uptake by primary producers in all assessed regions. This drawdown, however, is overestimated in May and June. At 100m, modelled Nitrate concentrations are consistently below the climatology in the Norwegian Sea (Figure 5). It is important to interpret this result with caution, considering the significant differences in data availability between winter and summer, particularly concerning the WOA2013 dataset. The same argument applies to the results shown in Figure 4 as the samples used for validation were predominantly obtained between spring and autumn, particularly in the Barents Sea where sampling usually starts in late spring. A direct comparison between the model and observations shows a negative bias in the model (values from -1.2 to -1.7; Table 5), when a full year and the entire water column are considered. The negative bias is more pronounced during the spring bloom period (April – June), and can be attributed to excessive nitrate consumption by phytoplankton in the model, resulting in a more negative bias towards the surface layers. The negative bias is observed to decrease during the summer months, coinciding with a seasonal decline in nitrate concentrations (Figure 5).

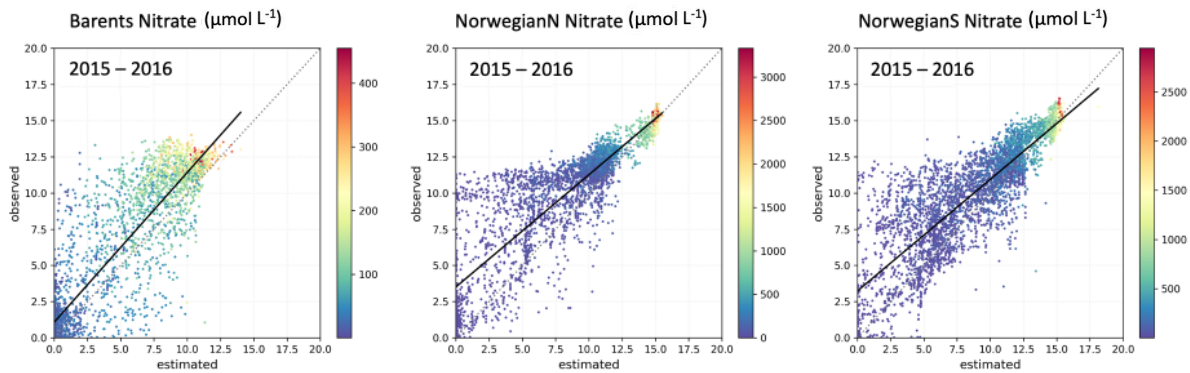

Figure 4: Co-located modelled (marked as "estimated" in the figure; x-axis) and the observed nitrate (µmol N/L; y-axis) for the subregions and regions in Figure 1b. The colours indicate the depth of sampling in meters. The model profiles are interpolated onto the sampling depths. The black line represents the regression line, while the dashed grey diagonal represents a perfect match.

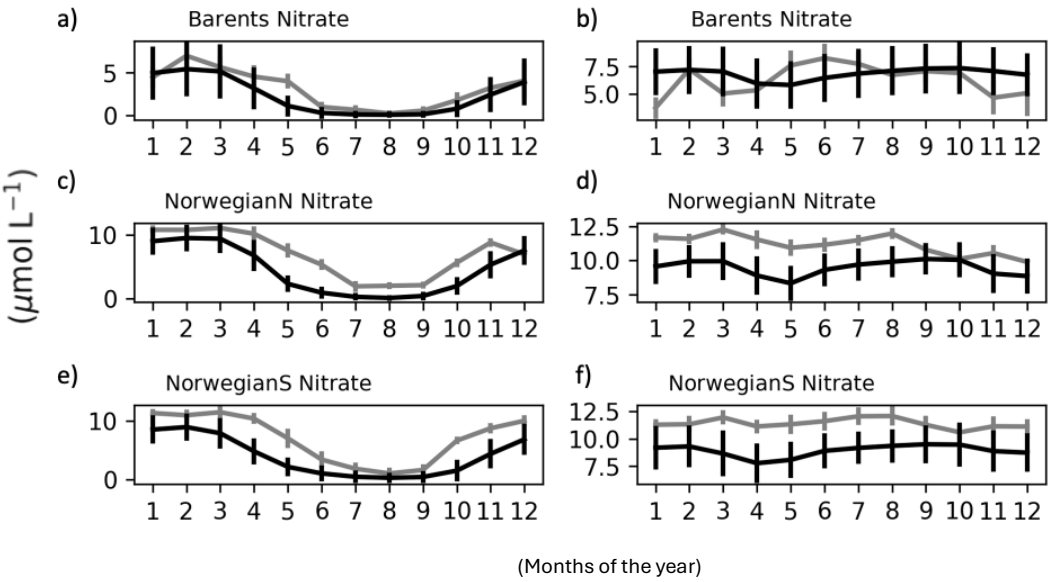

Figure 5: Seasonal cycle for nitrate at depths of 5 m (a, c, e) and 100 m (b, d, f). Model (black lines) and the WOA2013 (grey lines) regional monthly averages in selected areas (Barents, NorwegianS, and NorwegianN). The vertical lines represent the standard deviations of the model and WOA2013 data sets. The x-axis shows the months of the year (1: January; 12: December) and the y-axis shows the nitrate concentrations (µmol N/L).

Table 5: Estimated RMSD and bias for ARC BGC23 model predictions of nitrate (µmol N/L) for 2015 and 2016, compared to IMR2018 data. The RMSD for ARC BGC22 is shown in parentheses. The best value is highlighted in bold. In addition to the full-year metrics, model statistics are also provided for spring (April – June) and summer (July – September) periods and for different depth intervals with "0m – Bottom" covering the entire water column.

Depth (m)	Barents Full Year	NorwegianN Full Year	NorwegianS Full Year	Barents Spring	NorwegianN Spring	NorwegianS Spring	Barents Summer	NorwegianN Summer	NorwegianS Summer
RMSD, in µmol N/L									
0 -10	**1.244** (1.623)	3.641 **(3.333)**	3.053 **(3.026)**	**3.421** (4.424)	5.942 **(5.332)**	**3.517** (3.776)	**0.808** (1.032)	**1.244** (1.623)	3.641 **(3.333)**
10 -20	**1.428** (1.747)	3.760 **(3.455)**	3.149 **(2.816)**	**4.193** (5.103)	6.126 **(5.568)**	3.278 **(3.181)**	**0.887** (1.023)	**1.428** (1.747)	3.760 **(3.455)**
20-50	2.871 **(2.743)**	3.676 **(3.380)**	3.317 **(3.152)**	**3.982** (4.962)	5.162 **(4.579)**	3.741 **(3.707)**	2.782 **(2.431)**	2.871 **(2.743)**	3.676 **(3.380)**
50 -100	3.076 **(2.680)**	2.509 **(1.965)**	2.584 **(2.279)**	2.917 **(2.738)**	3.125 **(1.985)**	2.750 **(2.606)**	3.217 **(2.673)**	3.076 **(2.680)**	2.509 **(1.965)**
100 - 200	2.631 **(2.066)**	2.190 **(1.535)**	2.042 **(1.817)**	1.847 **(1.720)**	2.672 **(1.445)**	1.890 **(1.787)**	2.709 **(2.046)**	2.631 **(2.066)**	2.190 **(1.535)**
0 - Bottom	2.470 **(2.244)**	2.827 **(2.452)**	2.472 **(2.318)**	**3.167** (3.713)	3.808 **(3.209)**	**2.806** (2.817)	2.456 **(2.058)**	2.470 **(2.244)**	2.827 **(2.452)**
BIAS, in µmol N/L									
0 -10	-0.546 **(-0.526)**	-2.226 **(-1.674)**	-1.312 **(-0.304)**	**-2.228** (-3.319)	-5.016 **(-3.637)**	-1.777 **(-1.228)**	-0.352 **(-0.268)**	-0.546 **(-0.526)**	-2.226 **(-1.674)**
10 -20	**-0.629** (-0.630)	-2.475 **(-1.884)**	-1.523 **(-0.502)**	**-2.961** (-4.196)	-5.337 **(-3.805)**	-1.673 **(-1.097)**	-0.404 **(-0.324)**	**-0.629** (-0.630)	-2.475 **(-1.884)**
20 -50	-0.814 **(-0.660)**	-2.312 **(-1.592)**	-1.885 **(-1.111)**	**-2.775** (-4.063)	-4.423 **(-3.099)**	-2.258 **(-1.689)**	-0.629 **(-0.341)**	-0.814 **(-0.660)**	-2.312 **(-1.592)**
50 - 100	-1.575 **(-0.729)**	-1.509 **(-0.580)**	-1.429 **(-0.394)**	-1.164 **(-0.438)**	-2.646 **(-0.977)**	-1.738 **(-0.695)**	-1.685 **(-0.727)**	-1.575 **(-0.729)**	-1.509 **(-0.580)**
100 - 200	-1.980 **(-0.688)**	-1.619 **(-0.548)**	-1.125 **(0.047)**	-0.837 **(0.649)**	-2.267 **(-0.699)**	-1.087 **(0.261)**	-2.113 **(-0.764)**	-1.98 **(-0.688)**	-1.619 **(-0.548)**
0 - Bottom	-1.252 **(-0.553)**	-1.713 **(-0.941)**	-1.192 **(-0.353)**	-1.853 **(-1.838)**	-2.683 **(-1.520)**	-1.474 **(-0.661)**	-1.212 **(-0.433)**	-1.252 **(-0.553)**	-1.713 **(-0.941)**

IV.3 Phosphate

The quality of the product for phosphate was evaluated based on a comparison of the same databases (IMR2018 and WOA2013) used for nitrate (section IV.2) and applying the same techniques for the assessment. Co-located sample comparisons are presented in Figure 6, while comparisons to general monthly nutrient settings can be found in Figure 7. Modelled phosphate concentrations for the period 2015 - 2019 are generally slightly below climatological values, both at 5 m and 100 m. Co-located statistics demonstrate a slightly negative bias (approx. -0.1; Table 6). As observed with nitrate, the negative model bias is more pronounced during the spring bloom period. However, the magnitude of the bias for phosphate is smaller than that observed for nitrate. Similar to nitrate, monthly mean phosphate concentrations at 100 m in the Norwegian Sea are underestimated. Furthermore, there is a significant drawdown of phosphate during months 4 and 5 (Figure 7).

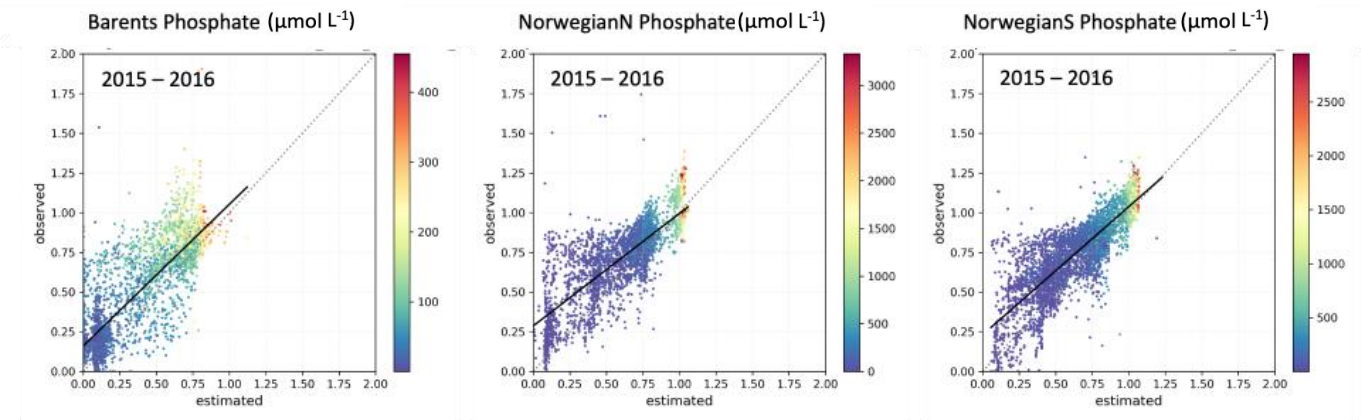

Figure 6: Co-located modelled (marked as "estimated" in the figure, x-axis) and the observed phosphate (µmol P/L; y-axis) for the subregions in Figure 1b. The colours indicate the depth of sampling in meters. The model profiles are interpolated onto the sampling depths. The black line represents the regression line, while the dashed grey diagonal represents a perfect match.

Table 6: Estimated RMSD and bias for ARC BGC23 model predictions of phosphate (µmolP/L) for 2015 and 2016, compared to IMR2018 data. The RMSD for ARC BGC22 is shown in parentheses. The best value is highlighted in bold. In addition to the full-year metrics, model statistics are also provided for spring (April – June) and summer (July – September) periods and for different depth intervals with "0m – Bottom" covering the entire water column.

Depth (m)	Barents Full Year	NorwegianN Full Year	NorwegianS Full Year	Barents Spring	NorwegianN Spring	NorwegianS Spring	Barents Summer	NorwegianN Summer	NorwegianS Summer
RMSD, in µmol P/L									
0-10	**0.142** (0.169)	**0.250** (0.264)	**0.214** (0.238)	**0.267** (0.380)	**0.396** (0.408)	**0.257** (0.319)	**0.124** (0.131)	**0.13** (0.162)	**0.083** (0.116)
10-20	**0.166** (0.188)	1.000 (1.003)	**0.228** (0.243)	**0.311** (0.421)	**0.395** (0.409)	**0.248** (0.292)	**0.128 (0.130)**	**1.618 (1.619)**	**0.243** (0.285)
20-50	0.233 (**0.216**)	**0.238** (0.252)	**0.732** (0.740)	**0.291** (0.403)	**0.331** (0.348)	**0.272** (0.315)	0.235 **(0.193)**	**0.178** (0.190)	**0.258** (0.287)
50-100	0.244 **(0.209)**	0.227 **(0.219)**	**0.173** (0.177)	**0.204 (0.203)**	0.191 **(0.163)**	**0.193** (0.217)	0.261 **(0.217)**	0.323 **(0.317)**	0.154 **(0.147)**
100-200	0.219 **(0.183)**	0.145 **(0.134)**	**0.214 (0.214)**	0.168 **(0.116)**	0.185 **(0.153)**	**0.146 (0.148)**	0.232 **(0.191)**	0.124 **(0.118)**	0.158 **(0.143)**
0 – Bottom	0.210 **(0.193)**	**0.356** (0.358)	**0.362** (0.367)	**0.250** (0.310)	**0.256 (0.255)**	**0.208** (0.240)	0.213 **(0.181)**	**0.566 (0.567)**	**0.176** (0.182)
BIAS, in µmol P/L									
0-10	**-0.092** (-0.118)	**-0.153** (-0.157)	**-0.116** (-0.139)	**-0.229** (-0.351)	-0.339 **(-0.298)**	**-0.181** (-0.242)	**-0.076** (-0.092)	**-0.097** (-0.139)	**0.008** (-0.081)
10-20	**-0.097** (-0.124)	**-0.223** (-0.226)	**-0.132** (-0.157)	**-0.264** (-0.395)	-0.348 **(-0.301)**	**-0.179** (-0.237)	**-0.078** (-0.095)	**-0.275** (-0.313)	**-0.096** (-0.178)
20-50	**-0.107** (-0.126)	-0.144 **(-0.136)**	**-0.165** (-0.185)	**-0.244** (-0.371)	-0.283 **(-0.242)**	**-0.199** (-0.245)	**-0.098** (-0.104)	**-0.089** (-0.098)	**-0.168** (-0.201)
50-100	-0.126 **(-0.098)**	**-0.088** (-0.063)	**-0.100 (-0.096)**	-0.145 **(-0.134)**	-0.167 **(-0.099)**	**-0.145 (-0.148)**	-0.136 **(-0.098)**	-0.0880 **(-0.072)**	-0.105 **(-0.099)**
100-200	-0.147 **(-0.089)**	-0.083 **(-0.046)**	-0.087 **(-0.06)**	-0.142 **(-0.074)**	-0.143 **(-0.079)**	-0.097 **(-0.067)**	-0.160 **(-0.098)**	-0.072 **(-0.038)**	-0.126 **(-0.103)**
0 - Bottom	-0.117 **(-0.099)**	-0.113 **(-0.094)**	**-0.105 (-0.102)**	**-0.195** (-0.233)	-0.184 **(-0.140)**	**-0.141** (-0.154)	-0.115 **(-0.090)**	**-0.106 (-0.103)**	**-0.116** (-0.126)

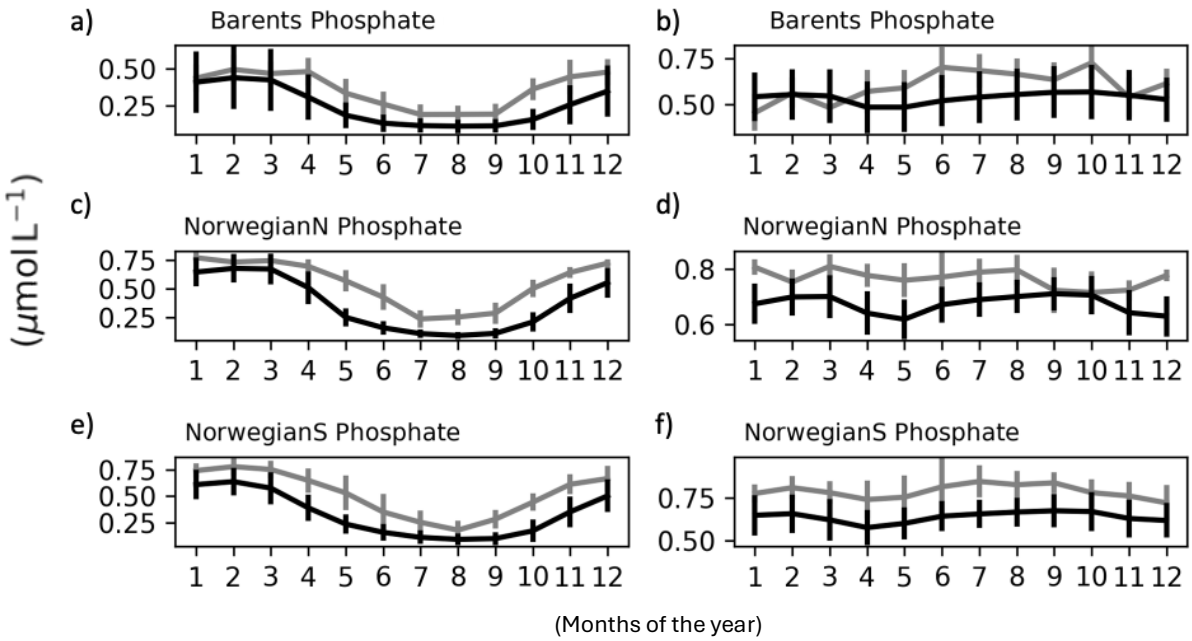

Figure 7: Seasonal cycle for phosphate at depths of 5 m (a, c, e) and 100 m (b, d, f). Model (black lines) and the WOA13 (grey lines) regional monthly averages in selected areas (Barents, NorwegianS, and NorwegianN). The vertical lines represent the standard deviations of the model and WOA13 data sets. The x-axis shows the months of the year and the y-axis shows the phosphate concentrations (µmol P/L).

IV.4 Silicate

The quality of the silicate product has been evaluated using the same comparative analysis and databases (IMR and WOA2013) as those applied for nitrate and phosphate (sections IV.2 and IV.3, respectively). Figure 8 presents co-located sample comparisons, while Figure 9 depicts comparisons against general monthly nutrient settings. The modelled silicate concentrations for the period between 2015 and 2019 are generally in good agreement with the climatology at 5 m and 100 m, with low co-located sample point biases ranging from 0.28 to 0.63. It should be noted that the model does not permit silicate uptake by diatoms when ambient silicate concentrations fall below 1 µmol Si/L. This limitation was implemented to improve the representation of silicate and primary production (Yumruktepe et al., 2022). The effect of this uptake threshold is evident in Figure 8, where silicate concentrations consistently remain above 1 µmol Si/L. The model depicts an anomalous decline in silicate levels in April, which is anticipated given the higher abundance of diatoms (responsible for silicate uptake) during the initial stages of year (model diatom results not shown).

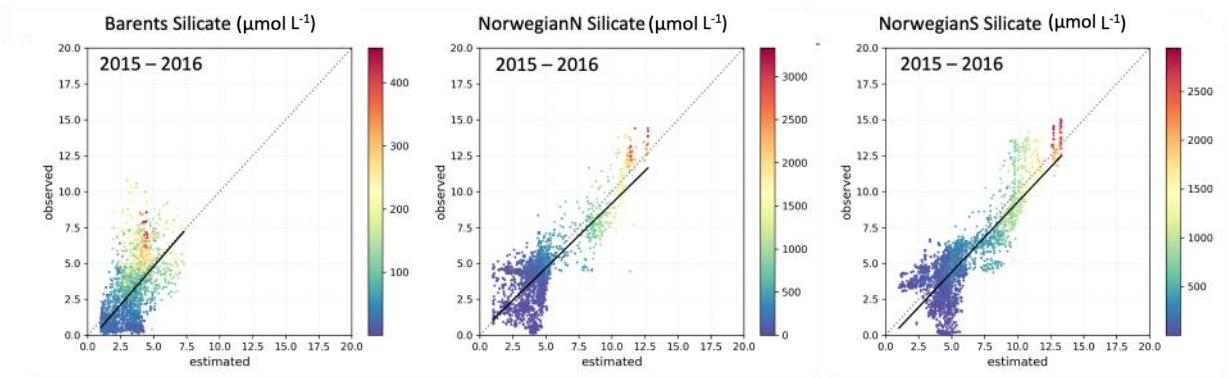

Figure 8: Co-located modelled (marked as "estimated" in the figure, x-axis) and the observed silicate (µmol Si/L; y-axis) for the subregions regions in Figure 1b.The colours indicate the depth of sampling in meters. The model profiles are interpolated onto the sampling depths. The black line represents the regression line, while the dashed grey diagonal represents a perfect match.

Table 7: Estimated RMSD and bias for ARC BGC23 model predictions of silicate (µmol Si/L) for 2015 and 2016 compared to IMR2018 data. The RMSD for ARC BGC22 is shown in parentheses. The best value is highlighted in bold. In addition to the full-year metrics, model statistics are also provided for spring (April – June) and summer (July – September) periods and for different depth intervals with "0m – Bottom" covering the entire water column.

Depth (m)	Barents Full Year	NorwegianN Full Year	NorwegianS Full Year	Barents Spring	NorwegianN Spring	NorwegianS Spring	Barents Summer	NorwegianN Summer	NorwegianS Summer
RMSD, in µmol Si/L									
0-10	1.888 **(0.850)**	1.943 **(1.666)**	2.236 **(1.780)**	**1.209** (1.312)	**1.580** (2.408)	**1.592** (2.072)	1.838 **(0.815)**	2.455 **(1.166)**	4.418 **(2.024)**
10-20	1.800 **(0.846)**	1.912 **(1.622)**	2.070 **(1.716)**	**1.265** (1.513)	**1.514** (2.430)	**1.532** (2.059)	1.741 **(0.78)**	2.371 **(1.057)**	3.790 **(1.587)**
20-50	1.551 **(1.144)**	**1.635** (1.722)	1.794 **(1.765)**	**1.189** (1.535)	**1.469** (2.378)	**1.328** (2.238)	1.475 **(1.114)**	1.896 **(1.172)**	2.550 **(1.47)**
50-100	**1.106** (1.464)	**1.078** (1.149)	**1.346** (1.522)	**0.969** (0.991)	**1.074** (1.320)	**1.039** (1.750)	**1.065** (1.469)	**0.824** (0.857)	1.008 **(0.898)**
100-200	**1.591** (1.643)	**0.724** (1.002)	**0.930** (1.296)	0.949 **(0.783)**	**0.789** (0.946)	**0.797** (1.359)	1.663 **(1.629)**	**0.566** (0.865)	**0.531** (0.680)
0 - Bottom	1.659 **(1.31)**	1.398 (1.396)	1.590 (1.588)	**1.087** (1.256)	**1.292** (1.766)	**1.243** (1.845)	1.663 **(1.300)**	1.606 **(1.031)**	2.216 **(1.391)**
BIAS, in µmol Si/L									
0-10	1.546 **(0.436)**	0.839 **(-0.138)**	1.253 **(-0.141)**	**0.704** (-0.989)	**-0.520** (-1.653)	**0.929** (-1.497)	1.504 **(0.551)**	1.962 **(0.492)**	4.386 **(1.944)**
10-20	1.451 **(0.408)**	0.812 **(-0.185)**	1.172 **(-0.286)**	**0.539** (-1.148)	**-0.513** (-1.642)	**0.910** (-1.434)	1.414 **(0.527)**	1.834 **(0.387)**	3.697 **(1.227)**
20-50	1.051 **(0.441)**	0.448 **(-0.251)**	0.932 **(-0.203)**	**0.323** (-1.196)	**-0.706** (-1.568)	**0.459** (-1.584)	0.988 **(0.587)**	1.271 **(0.420)**	2.165 **(0.800)**
50-100	**0.119** (0.782)	**0.188** (0.373)	0.446 **(0.160)**	**-0.097** (-0.437)	**-0.494** (-0.151)	**0.007** (-0.636)	**0.019** (0.912)	**0.132** (0.410)	0.484 **(0.208)**
100-200	**-0.617** (1.031)	**-0.083** (0.593)	**0.241** (0.758)	-0.527 **(0.288)**	-0.461 **(0.411)**	**0.001** (0.600)	**-0.748** (1.054)	**-0.184** (0.549)	**-0.012** (0.334)
0 - Bottom	**0.335** (0.667)	0.283 **(0.234)**	0.633 **(0.277)**	**0.061** (-0.484)	**-0.314** (-0.343)	**0.395** (-0.486)	**0.249** (0.756)	0.677 **(0.469)**	1.260 **(0.721)**

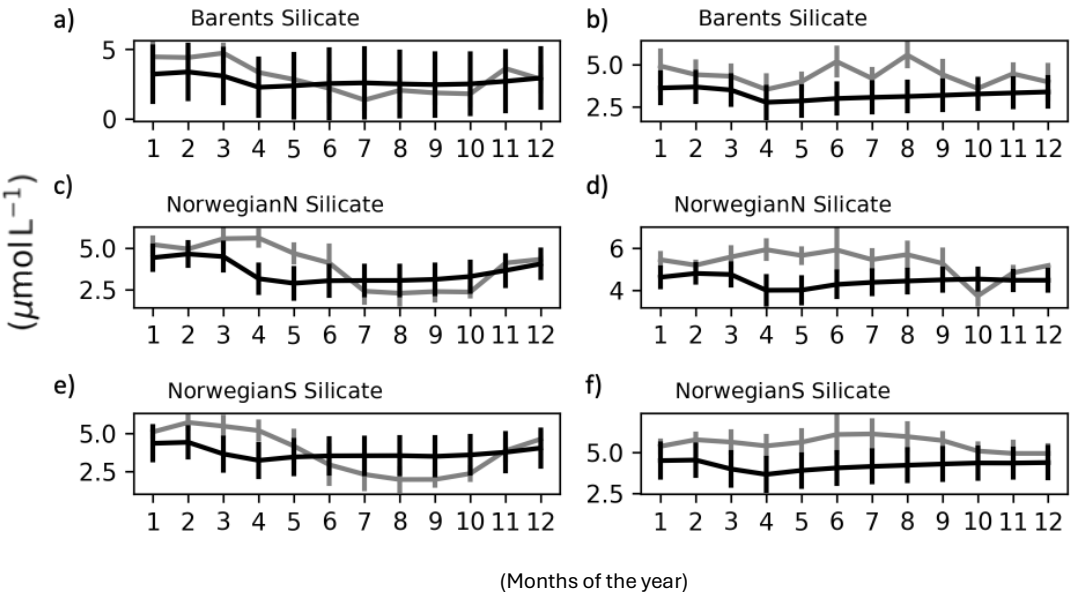

Figure 9: Seasonal cycle for silicate at depths of 5 m (a, c, e) and 100 m (b, d, f). Model (black lines) and the WOA2013 (grey lines) regional monthly averages in selected areas (Barents, NorwegianS, and NorwegianN). The vertical lines represent the standard deviations of the model and WOA2013 data sets. The x-axis shows the months of the year and the y-axis shows the silicate concentrations (µmol Si/L).

IV.5 Surface dissolved inorganic carbon

The quality of the product for surface-dissolved inorganic carbon has been evaluated by comparison with a reprocessed time series of monthly global reconstructed surface ocean pCO2, air-sea fluxes of CO2 L4 product (https://doi.org/10.48670/moi-00047). The model and observed data are compared using monthly and regionally averaged time series (Figure 10). Overall, the model results reproduce the observed seasonality and show good agreement based on statistical metrics (Table 1), although the model overestimates mid-year concentrations of CO_2 in the Barents Sea.

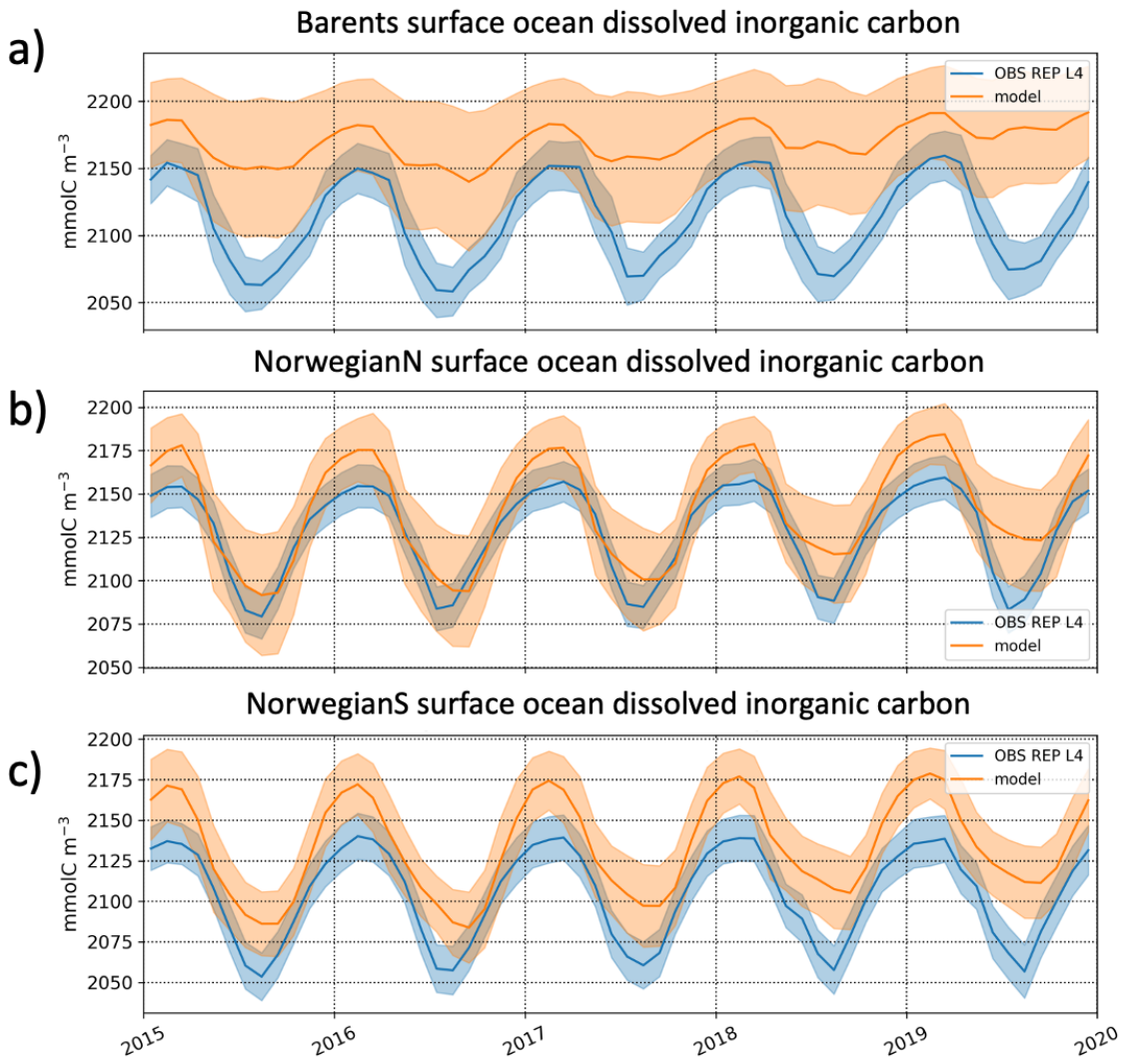

Figure 10: Monthly and regionally averaged surface dissolved inorganic carbon concentrations for the period between 2015 and 2019. The model results (orange) are averaged within the region boundaries (see Figure 1b). The observed data (blue) are retrieved as monthly averaged reprocessed L4 product MULTIOBS_GLO_BIO_CARBON_SURFACE_REP_015_008

IV.6 Surface CO2 flux

The quality of the product for the surface downward flux of inorganic carbon has been evaluated through a comparison with a reprocessed time series of monthly global reconstructed surface ocean pCO2, air-sea fluxes of CO2 L4 product (https://doi.org/10.48670/moi-00047). A comparison between the model and observed data is made using monthly and regionally averaged time series (Figure 11). The model results indicate an overestimation of CO_2 flux against the validation data provided as documented with the statistics (Table 1). This issue will be addressed with particular attention in future iterations of the model.

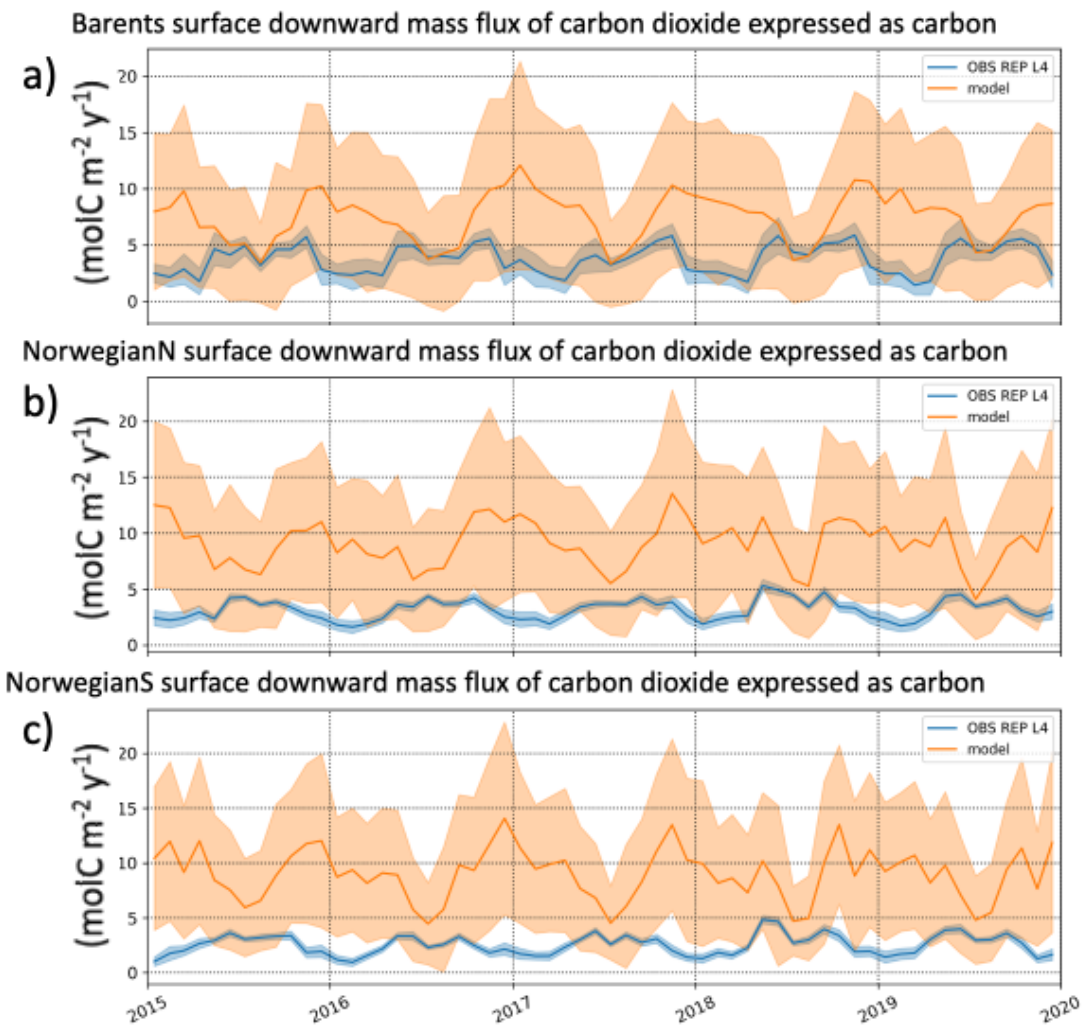

Figure 11: Monthly and regionally averaged surface downward flux of inorganic carbon for the period between 2015 and 2019.The model results (orange) are averaged within the region boundaries (see Figure 1b). The observed data (blue) are retrieved as monthly averaged reprocessed L4 product MULTIOBS_GLO_BIO_CARBON_SURFACE_REP_015_008.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system. In includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality (Table 8).

Table 8: List of events and changes for product ARCTIC_ANALYSISFORECAST_BGC_002_004

Date	Change/Event description	System version	Other
28/05/2025	Advection of nutrients and algae in sea ice.	3.4	
30/05/2024	A module representing the algae growing from the bottom of the sea ice has been included.	3.3	
30/06/2023	Coccolithophores were added as a new phytoplankton functional type state variable. As coccolithophores are responsible for CaCO3 production, CaCO3 was also added as a new state variable, incorporated to water column and sediment layer. Detritus sinking speed is changed from a constant value to a variable one allowing the large/small plankton to increase/decrease in their respective ratios to the production. Chlorophyll-a was included in the growth, term modifying the light-limitation, thus increasing its influence on productivity., As a result, under high chlorophyll-a concentrations, phytoplankton become more productive taking advantage of high light absorption. These changes required substantial tuning of the parameters. To refine the parameter set, BGC-Argo data from the Nordic Seas were used, particularly chlorophyll-a and oxygen estimates provided by the BGC-Argo floats.	3.2	
29/11/2022	The algorithm used for nudging of Chlorophyll-a data was changed, from Uitz et al. (2006) to Ardyna et al. (2013). The new approach leverages an Arctic-focused formulation and parameterization.	3.1	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

The lack of validation data dictated that we present multiple versions of the model. This section aims to provide a short summary of the main changes in Product Quality between the previous and current version.

Version 3.4

The model has been updated to include advection of nutrients and algae in sea ice.

Since there is no satellite or in situ measurements below sea ice, the validation results presented are not affected by this change. In this context, nutrients, chl-a, CO_2 concentration and flux statistics refer to ARC23 (The lack of validation data dictated that we present multiple versions of the model).

Version 3.3

A module representing the algae growing from the bottom of the sea ice has been incorporated into the model.

The related biomass is accounted for in the top 2 m of the water column. However, since there is no satellite or in situ measurements below sea ice, the validation results presented are not affected by this change.

Version 3.2

Compared to the previous version, 3.1, the chlorophyll-a bias has been significantly reduced for the 1-day forecast range during the spring bloom period. During the summer season, the model demonstrates greater variability in chlorophyll-a bias, exhibiting oscillating values around the zero-bias line. This behaviour is primarily due to the incorporation of chlorophyll-a into the growth term, which allows a higher growth rate during low-light conditions and consequently reduces the negative bias. Due to the substantial alterations to the model code, additional parameter tuning is required to achieve more precise estimation of chlorophyll-a during the summer season.

The validation results for nutrients exhibit a combination of outcomes when compared to those from the previous product version, 3.1. The overall statistics for V3.1 and V3.2 differ much less than the variability in the time series of the metrics. With further parameter tuning, we expect future iterations of the model to deliver measurable improvements in performance.

This is the first time we included CO_2 statistics. The model is capable of representing surface CO_2 well, though the downward flux of CO_2 requires further attention in future versions of the model.

VII REFERENCES

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